

Cross-Workspace Compliance Investigation

Industry: Financial Services — Hedge Funds, Asset Managers, Broker-Dealers

Feature: Workspace Bridges **Platform:** Roundtable by Foxtrot Communications

Executive Summary

In multi-desk financial firms, compliance investigations require data that spans trading systems, risk platforms, communication archives, and regulatory databases — each owned by different teams. Today, this coordination happens through emails, Slack threads, and manual data requests, producing no unified audit trail.

Roundtable's **Workspace Bridges** enable AI-mediated cross-team collaboration where each team retains its own tools and data access, but can open a shared, auditable channel for joint investigation. Every message, tool call, and AI response is logged with workspace attribution for SEC/FINRA compliance.

The Problem

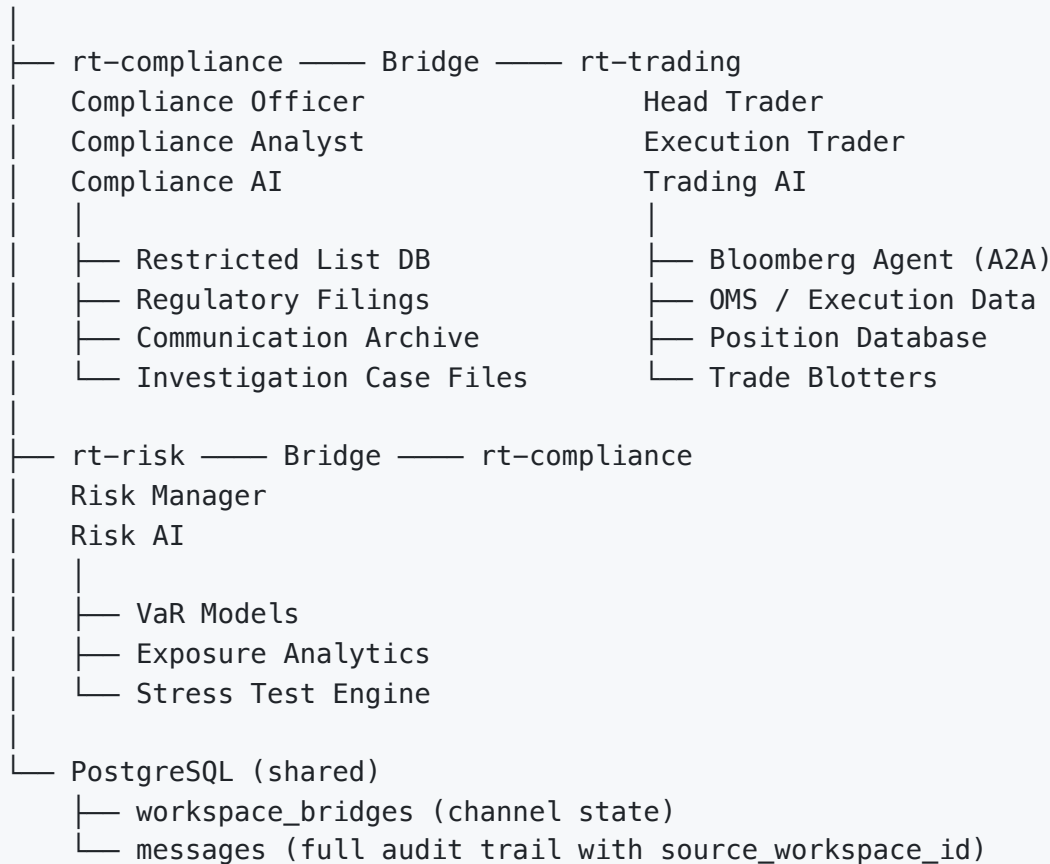
When a compliance analyst spots an anomalous securities transaction, they need:

1. **Trading data** from the execution desk (fills, timestamps, counterparties)
2. **Position data** from the portfolio management system
3. **Communication records** for the relevant trader
4. **Restricted list** checks against their own compliance database

Each of these data sources is owned by a different team, accessed through different tools, and protected by different access controls. The investigation requires coordination across organizational boundaries — without breaking information barriers.

Architecture

GKE Cluster (Fund Infrastructure)



Key Design Principles

- **Tool isolation:** The compliance AI cannot access Bloomberg directly. The trading AI cannot access the restricted list. Each workspace's AI retains only its own tools. Cross-workspace queries are explicit and logged.
- **Human-in-the-loop:** The bridge must be accepted by the target workspace. Both sides see every message. No silent data access.
- **Audit trail:** Every message in a bridge is tagged with `source_workspace_id`. SEC/FINRA examiners can reconstruct exactly what was asked, by whom, when, and what data informed the response.

Walkthrough: Investigating an Anomalous Trade

Step 1 — Compliance Detects an Anomaly

The compliance analyst queries their surveillance database in the `rt-compliance` workspace:

Compliance Analyst:

@ai Show me all trades flagged by our wash trade detection model in the past 48 hours.

Compliance AI → [queries surveillance database]

3 trades flagged:

Symbol	Side	Qty	Price	Time	Score
AAPL	Buy	5,000	198.42	May 14, 10:32a	0.87
AAPL	Sell	5,000	198.45	May 14, 10:34a	0.87
TSLA	Buy	2,000	178.90	May 14, 2:15p	0.62

The AAPL pair (buy/sell within 2 minutes, same quantity, \$0.03 spread) scores 0.87 on the wash trade model.

This warrants investigation.

Step 2 — Opening a Bridge to Trading

The compliance analyst needs execution details that only the trading desk has:

Compliance Analyst clicks: [Bridge → Trading Desk]

System: Bridge request sent to `rt-trading`.
Waiting for acceptance...

Head Trader (in `rt-trading`): [Accept Bridge]

System: Bridge established between Compliance and Trading Desk.
All messages in this channel are visible to both workspaces and logged for regulatory compliance.

Step 3 — Cross-Workspace AI Query

The compliance analyst queries the **trading workspace's AI**, which has access to Bloomberg and the order management system:

Compliance Analyst:

@trading-ai Show me the full execution details for the AAPL trades on May 14 between 10:30a and 10:35a. Include counterparties, venue, and order origination.

Trading AI → [queries OMS, Bloomberg execution data]

Order 1 (Buy):

Symbol: AAPL | Side: Buy | Qty: 5,000 | Avg: \$198.42
Venue: NASDAQ | Counterparty: Citadel Securities
Order Type: Limit | Originator: jsmith (Algo Desk)
Parent Order ID: PO-2024-88412
Strategy: VWAP

Order 2 (Sell):

Symbol: AAPL | Side: Sell | Qty: 5,000 | Avg: \$198.45
Venue: NYSE | Counterparty: Virtu Financial
Order Type: Market | Originator: jsmith (Algo Desk)
Parent Order ID: PO-2024-88415
Strategy: Manual

Note: Different parent order IDs suggest these were independent orders, not legs of the same strategy.
The buy was algorithmic (VWAP), the sell was manual.

Step 4 — Compliance Uses Own Tools

The compliance analyst uses **their own AI** for restricted list checks and communication review:

Compliance Analyst:

@ai Check if jsmith has any active restrictions on AAPL, and pull any communications mentioning AAPL from the past week.

Compliance AI → [queries restricted list, comms archive]

Restricted List Check:

- ✓ AAPL is NOT on the firm restricted list
- ✓ jsmith has no personal trading restrictions on AAPL

Communications (past 7 days):

3 results for "AAPL" in jsmith's communications:

1. May 13, 4:45p – Slack to mwilson (PM):
"AAPL looking heavy into close, might want to trim"
2. May 14, 10:28a – Bloomberg chat to external (Citadel):
"Working a 5k AAPL buy, VWAP, 30min horizon"
3. May 14, 10:33a – Slack to self-note:
"Filled on AAPL buy. PM says reverse it, wrong account"

The third communication is significant – "wrong account" suggests the sell was a correction, not a wash trade.

Step 5 — Escalation to Risk

Compliance opens a second bridge to the Risk workspace:

Compliance Analyst:

@risk-ai What's the current net AAPL exposure across all accounts managed by the equity desk?

Risk AI → [queries position and exposure systems]

AAPL Exposure Summary (as of May 15, 8:00a):

Account A (Equity Long/Short):	+12,000 shares (\$2.38M)
Account B (Market Neutral):	-3,000 shares (\$595K)
Account C (Algo Trading):	0 shares

Net Firm Exposure: +9,000 shares (\$1.79M)

Account C (Algo Trading, managed by jsmith) shows zero position. The 5,000 buy/sell cycle on May 14 resulted in a flat position – consistent with a trade correction.

Step 6 — Resolution

The compliance analyst has a complete picture across three workspaces:

- **Trading desk:** Independent orders with different strategies
- **Compliance tools:** Communication trail explaining the correction
- **Risk systems:** Net-zero position confirming no directional intent

Compliance Analyst:

@ai Summarize this investigation and draft a disposition memo for the AAPL wash trade alert (May 14).

Compliance AI → [generates memo with cross-workspace citations]

DISPOSITION: FALSE POSITIVE – Trade Correction

Summary: The AAPL buy/sell pair flagged on May 14 (score: 0.87) was a trade correction, not a wash trade. Evidence:

1. Different parent order IDs (P0-88412 vs P0-88415)
2. Different strategies (VWAP vs Manual)
3. Trader communication: "PM says reverse it, wrong account"
4. Net-zero position in the originating account

Recommendation: Close alert. No further action required.

Total investigation time: ~15 minutes (vs. 1-2 days with manual coordination).

Regulatory Compliance

Requirement	How Roundtable Addresses It
Books and records (SEC 17a-4)	Every bridged message is persisted with timestamp, user ID, workspace origin, and tool call details
Supervisory review (FINRA 3110)	Bridge requests require acceptance; all AI tool calls are logged and reviewable
Communication surveillance	AI queries across workspaces are captured as structured data, not ephemeral chat
Information barriers	Workspaces maintain tool isolation — compliance cannot directly access trading data without an explicit, logged bridge
Audit trail reconstruction	<code>source_workspace_id</code> on every message enables full reconstruction of cross-desk information flow

Technical Implementation

Database Schema

```
CREATE TABLE workspace_bridges (  
  id SERIAL PRIMARY KEY,  
  workspace_a TEXT NOT NULL REFERENCES workspaces(id),  
  workspace_b TEXT NOT NULL REFERENCES workspaces(id),  
  status TEXT NOT NULL DEFAULT 'pending'  
    CHECK(status IN ('pending', 'active', 'closed')),  
  opened_by INTEGER REFERENCES users(id),  
  accepted_by INTEGER REFERENCES users(id),  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  closed_at TIMESTAMP,  
  UNIQUE(workspace_a, workspace_b)  
);
```

API Endpoints

Method	Endpoint	Description
POST	/api/bridge/:target_workspace_id	Open a bridge request
PUT	/api/bridge/:bridge_id/accept	Accept a bridge request
PUT	/api/bridge/:bridge_id/close	Close a bridge
GET	/api/bridges	List active bridges
GET	/api/bridge/:bridge_id/messages	Get bridged conversation

Cross-Workspace AI Routing

When a user types `@trading-ai <query>`, the message routes to the target workspace's AI with the target's tools and system prompt. `@ai` always routes locally. Both responses are broadcast to all bridged users.

Deployment

For a 100-person fund with 5 desks:

```
GCP_PROJECT=fund-infra ./deploy-gke.sh compliance "Compliance"
GCP_PROJECT=fund-infra ./deploy-gke.sh trading "Trading Desk"
GCP_PROJECT=fund-infra ./deploy-gke.sh risk "Risk Management"
GCP_PROJECT=fund-infra ./deploy-gke.sh quant "Quant Research"
GCP_PROJECT=fund-infra ./deploy-gke.sh ops "Operations"
```

5 workspace pods, shared PostgreSQL, shared A2A agent pool. Each desk has its own AI, tools, and data access. Bridges connect them on demand.

Pricing

Tier	Workspaces	Bridges	
Open Source	1	—	Free
Team	Up to 5	Within team	Contact sales
Enterprise	Unlimited	Cross-workspace	Contact sales

Roundtable by [Foxtrot Communications](#) — AI workspace platform for teams that can't send their data to the cloud.